Modeling the Function of Visual Pathways:

Modeling the Effect of Learning on Dorsal Pathway Function in Autism

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Outline

01 Introduction

02 Purpose

03 Methods and Results

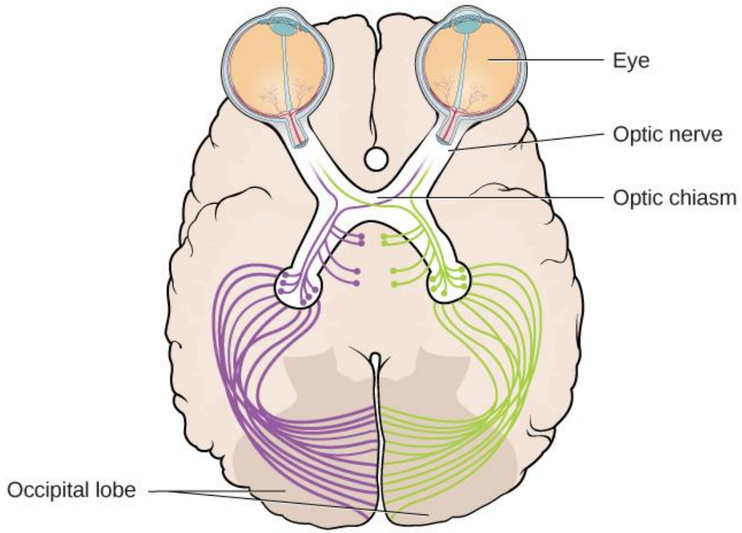
- Modeling Both Dorsal and Ventral Pathways
- Dorsal Stream Dysfunction and Vision Problems in Autism

04 Conclusion

Introduction

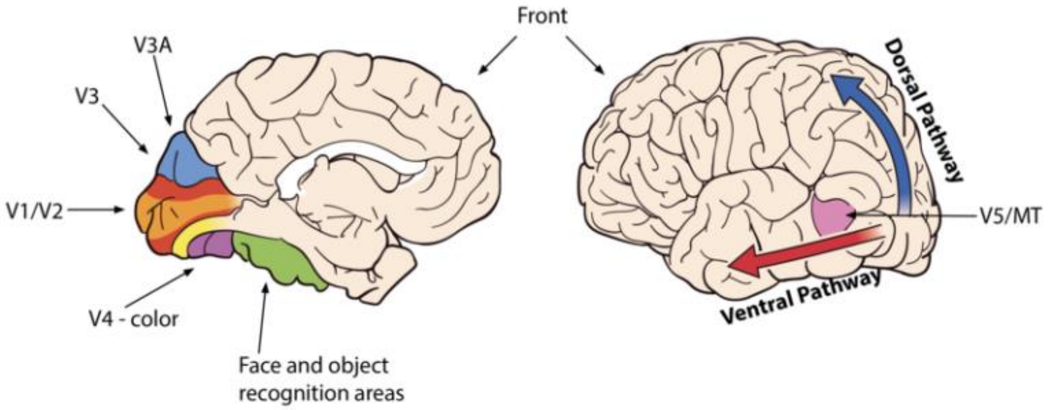
Visual Cortex

- The brain controls our thoughts, memory and speech, movements, and the function of many organs within our body.
- The **visual cortex** is the primary cortical region of the brain that receives, integrates, and processes visual information relayed from the retinas.



Visual Cortex Pathways

- Two main pathways:
  - *Ventral stream ("what" pathway):* Identifying and **recognizing** objects.
  - *Dorsal stream ("where" pathway):* Analyzing the movement and **spatial relationships** of those objects.



Related Works

- **Ventral pathway** modeling methods:
  - Goal-driven deep learning models
  - Unsupervised neural network models
  - GANs

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Purpose

The purpose of this study was to

To ***investigate roles of ventral and dorsal streams*** in healthy brain

To ***model dorsal and ventral streams*** in healthy brain

To investigate ***vision problems*** in autism spectrum disorder

To ***model*** dorsal and ventral streams in autism spectrum disorder

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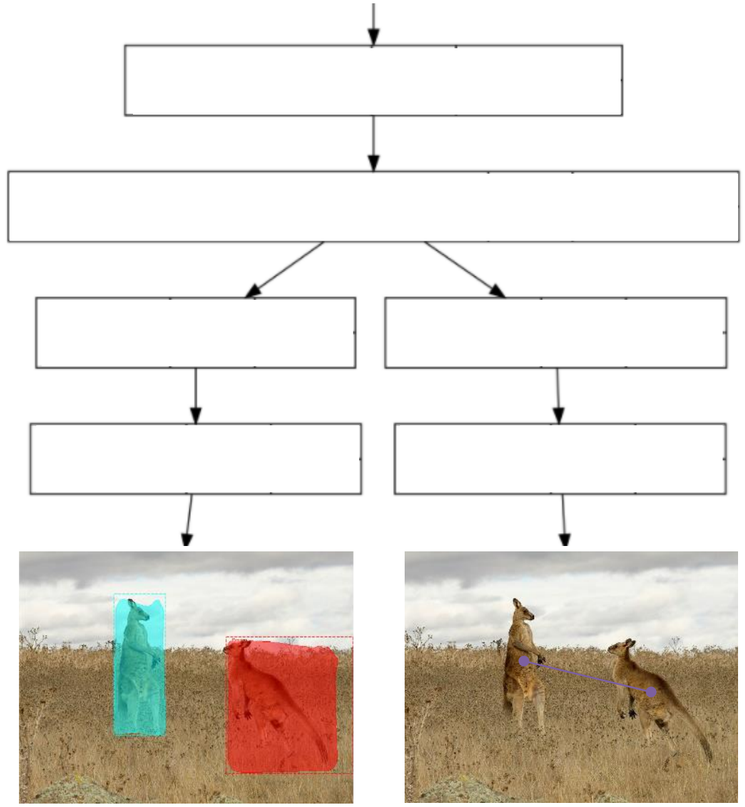
Methods

Modeling Both Dorsal and Ventral Pathway

Method

- Our final model consists of two parts:
  - Object detection
  - Distance estimation



Detection
Output: bounding boxes

Distance Estimation
Output: distance

Approaches

01

Custom Model

Building a model from scratch.

02

Modifying Mask-RCNN

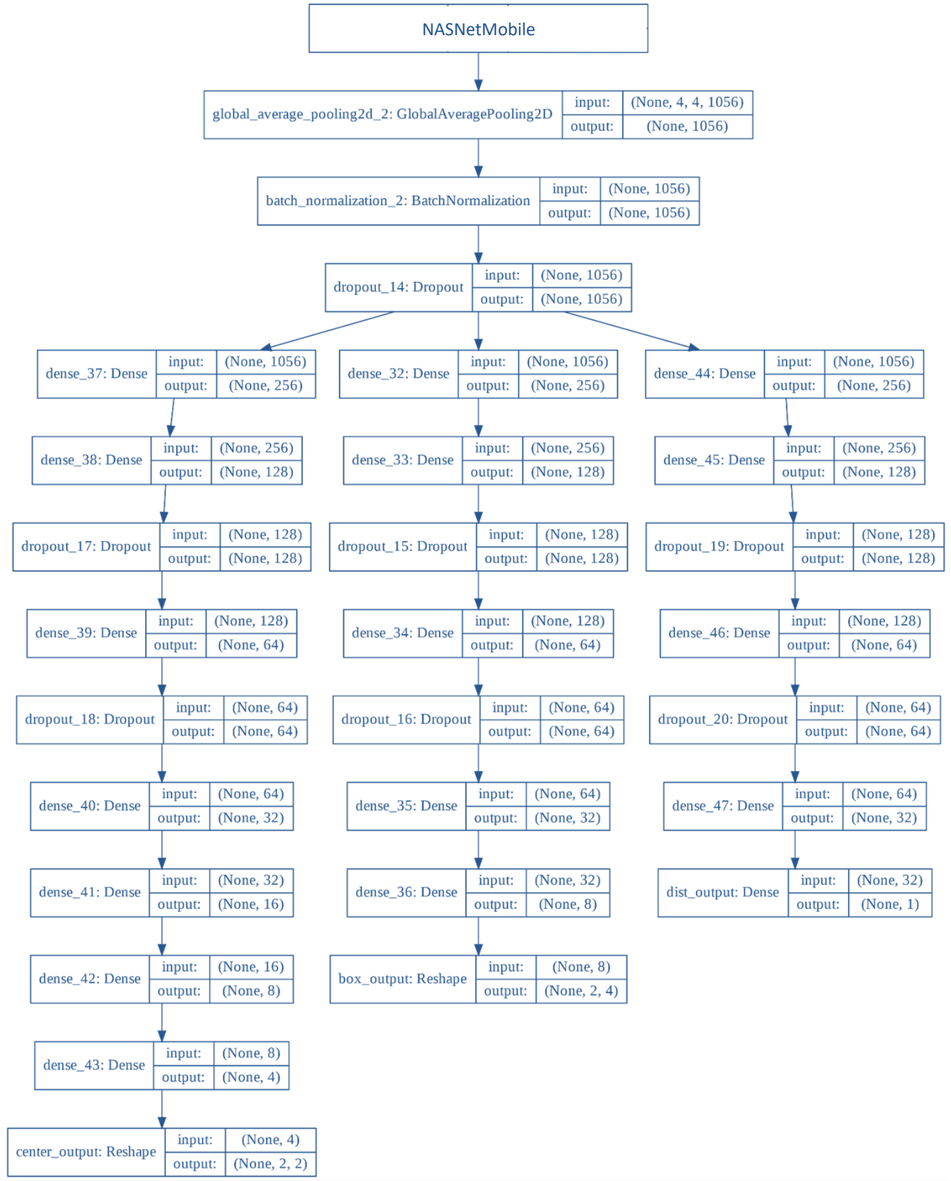
Adding distance estimation branch to Mask-RCNN.

1. Custom Model

Model: Data and Architecture

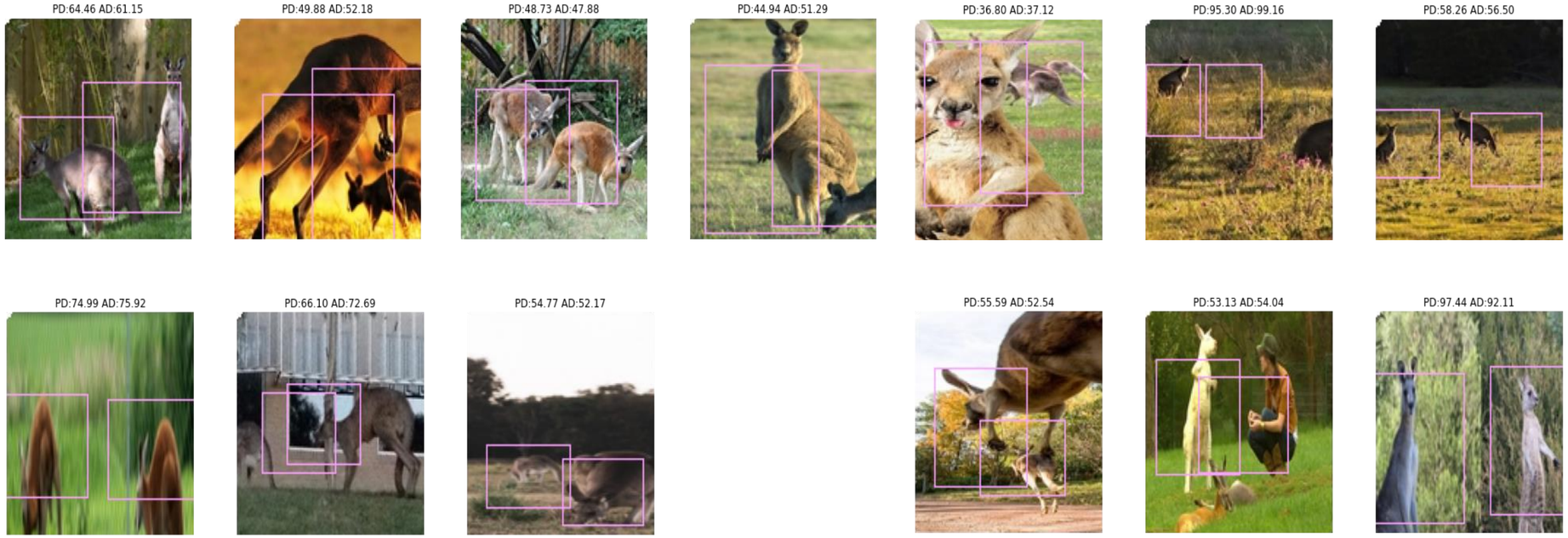
- Generated Data:
  - 202 Image
  - Each image containing exactly 2 kangaroos
  - XML annotation file

- Model: NASNetMobile + Multi-Output model



Model: Result

- Validation result:
  - *Distance Mean Absolute Error:* 9.57

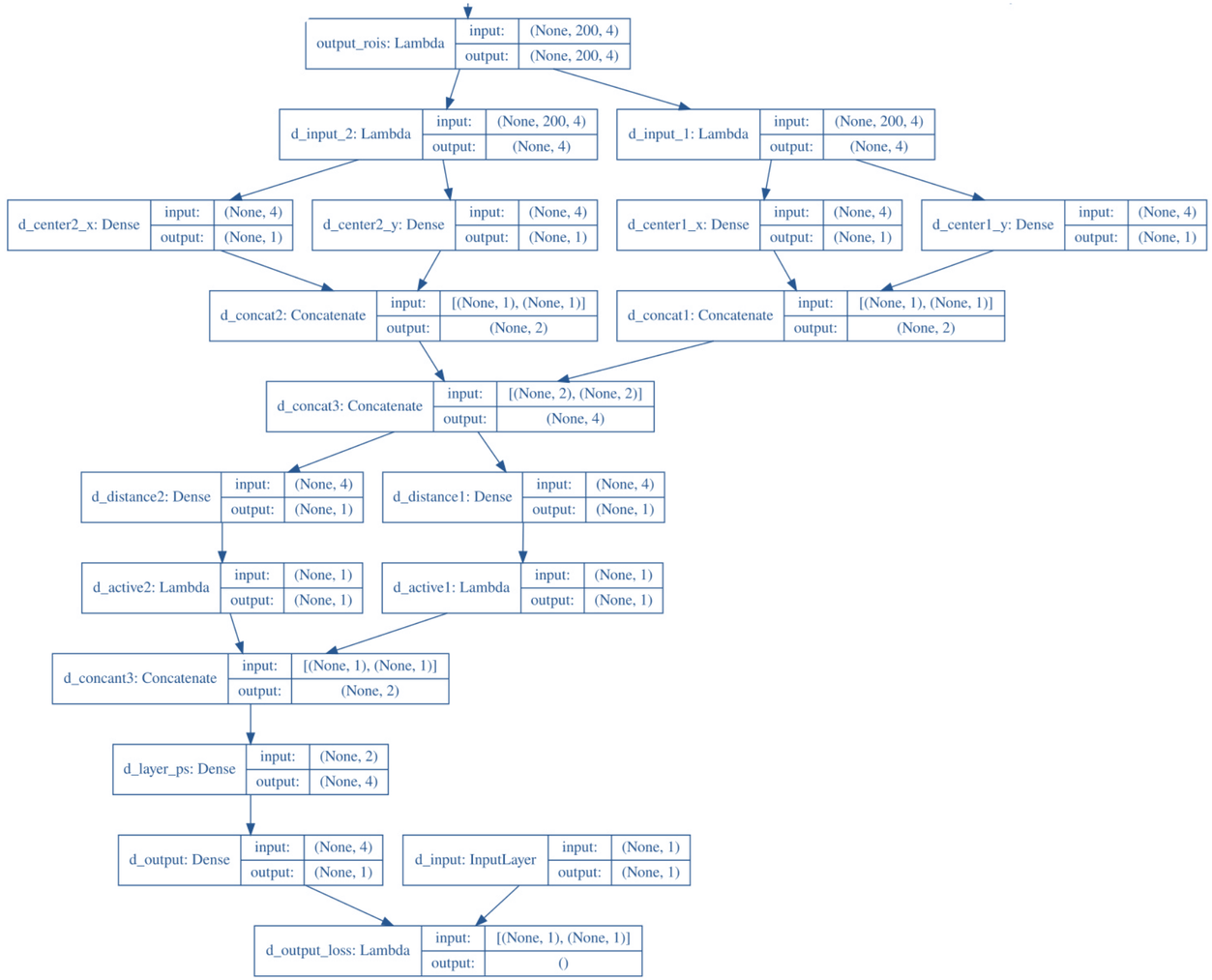


2. Modifying Mask-RCNN

Model: Data and Architecture

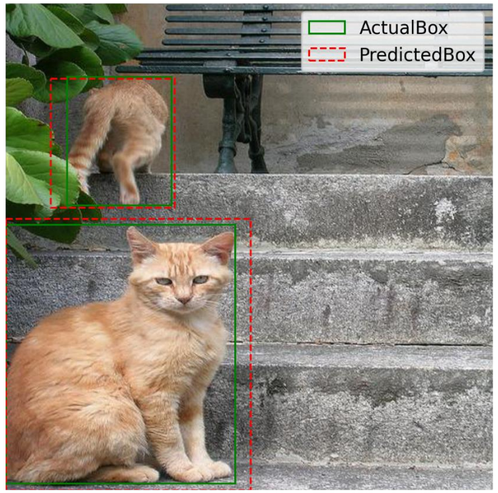
- Generated Data:
  - 266 Image (480 x 480)
  - Each image containing exactly 2 cats
  - XML annotation file

- Model: Mask-RCNN + Distance Branch

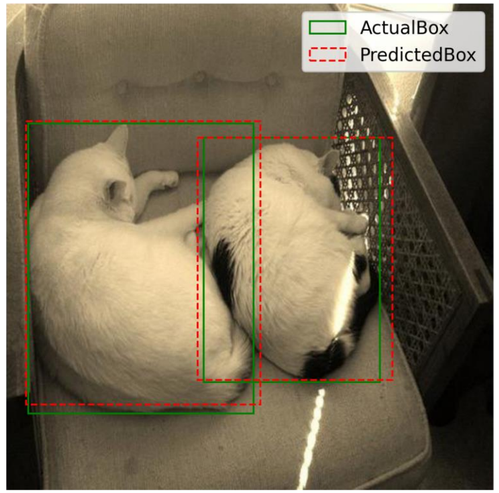


Model: Result

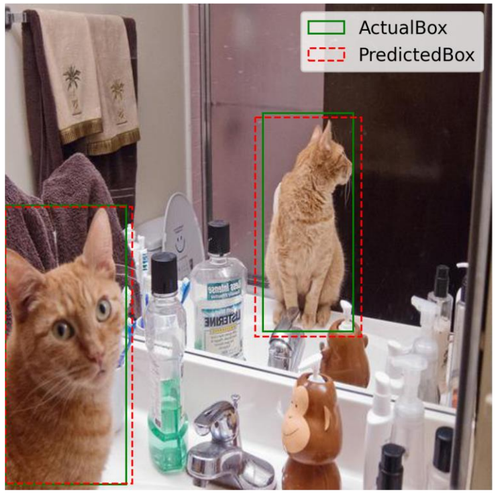
Actual:210.01 - Predicted: 202.99



Actual:149.24 - Predicted: 164.33



Actual:267.88 - Predicted: 234.61



Actual:219.00 - Predicted: 200.09

Modeling Dorsal Stream Function in Autism

Investigating the Impact of Learning on Vision in Autism

- **Learning** can help improve vision problems in autism.

- Modeling steps:
  - **Training the visual system model** to achieve acceptable accuracy in detecting objects and finding distance between them.
  - **Changing the weights of some layers** in the model that has been trained.
  - **Retraining** model.
  - Examine whether this training process has improved the performance of our model.

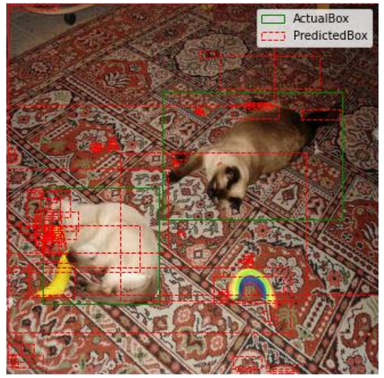
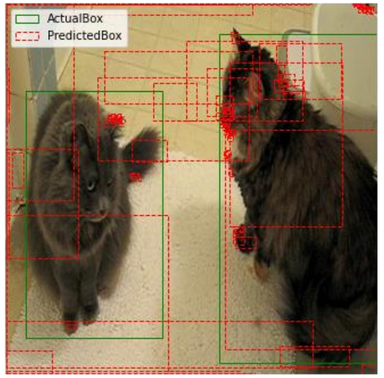
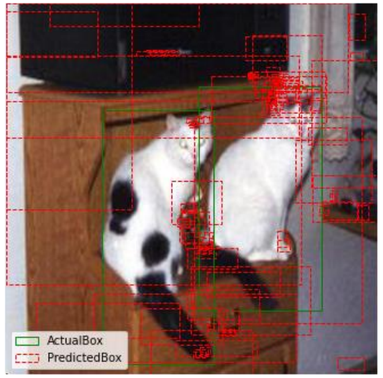
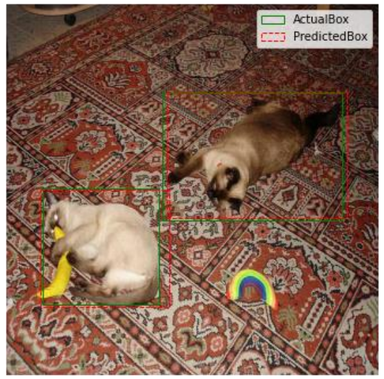
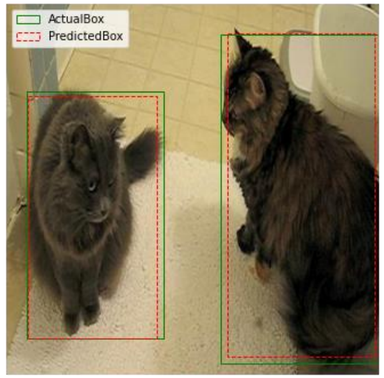
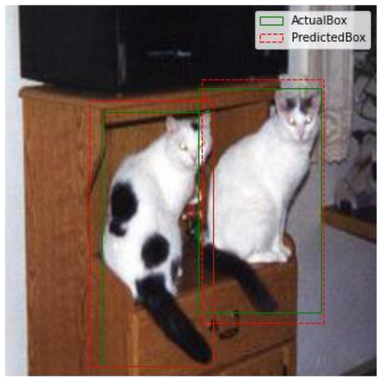
Changing the Weights of Layers: Details and Results

- **Replacing weights by 0:** There is no learning and even after 250 epochs all weights remain zero and the output distance is 0.

- **Replacing weights by 1:** The learning is so slow that after 250 epochs the change in weight values is very small and the output distance is still 0.

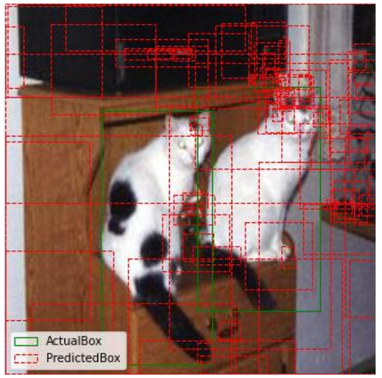
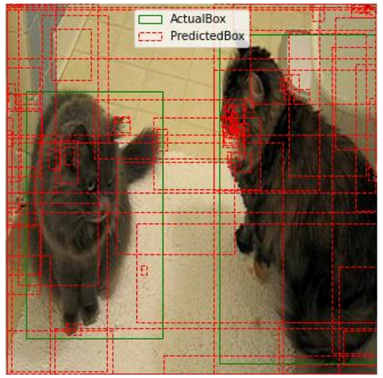
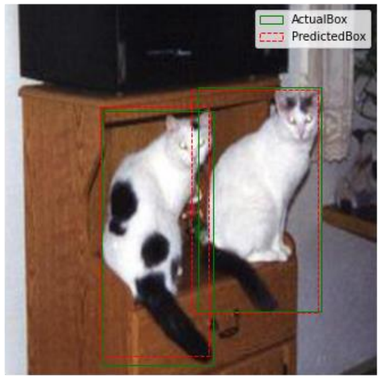
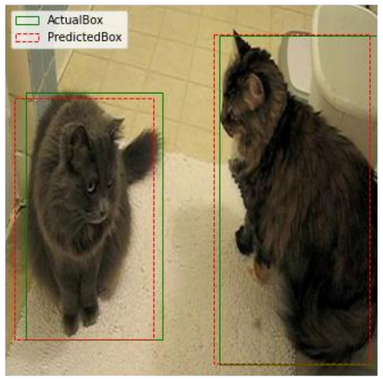
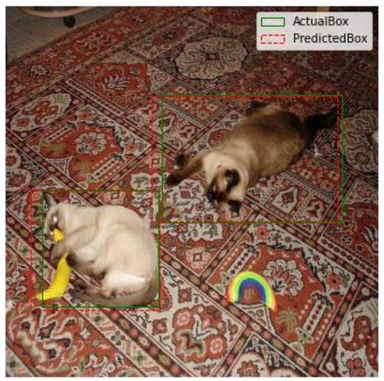
Changing the Weights of Layers: Details and Results

- **Replacing weights by random numbers:** For random numbers between 0 and 1, after 30 epochs:

Before Training	Actual:227.58 - Predicted: 814.12 	Actual:264.26 - Predicted: 884.82 	Actual:141.91 - Predicted: 900.82 
After Training	Actual:227.58 - Predicted: 156.36 	Actual:264.26 - Predicted: 172.62 	Actual:141.91 - Predicted: 165.25 

Changing the Weights of Layers: Details and Results

- **Addition of weights with a small number:** For adding 0.1 to all weights, after 30 epochs model can detect boxes and distance almost accurate.

Before Training	Actual:141.91 - Predicted: 233.24 	Actual:264.26 - Predicted: 773.03 	Actual:227.58 - Predicted: 430.96
After Training	Actual:141.91 - Predicted: 152.74 	Actual:264.26 - Predicted: 210.21 	Actual:227.58 - Predicted: 198.82 

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- We provided a model for both dorsal and ventral streams.
- We investigated the impact of learning on vision in autism.

- ***Future Works***
  - Presenting a **more biologically plausible** model using spiking neurons and better mapping of cortical areas.
  - Modeling **more functionalities** of the dorsal stream.
  - Modeling more **visual system dysfunctions**.